

Formulas and variables — reader’s companion

Canonical name: 21_Formulas_and_variables.md

Purpose: One place to decode formulas and symbols while reading #03 (and the rest of the stack) before #05. Restates each equation in plain English, defines variables, and notes where it lives in the repo.

Not a substitute for: 05_Model_Nesting_Note_v1.md (canonical theory) · 01_forward_plan_reading_guide.md (shorthand glossary).

Living Q&A: append misses to 20_feedback_questions_and_observations.md.

PDF (you run locally): ./scripts/convert_single_md_to_pdf.sh 3-Master_Plan/21_Formulas_and_variables.md "" pdf_styles_narrow.css

Compiled: 2026-06-16 from #03 – #07, #05, #12, Alex_model_interpreted.md, 538_Cell10_Generative_Manual.md, D10 export templates.

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Cross-domain column map (symbols $\rightarrow$ CSV columns)

Theory symbol	Plain English	Basketball column	Army column	Tenure column
a_i, A_i	Own ability / performance	perf (often PPM z within season)	Own talent bucket (TB) / controls	Own publication rate / controls
$\bar{a}_t$	Team / pool mean quality	team_mean (generative); empirical proxy = poolq_loo	Pool minus mean construct	Dept context (implicit in LOO)
L_Q	LOO mean teammate quality	poolq_loo	LOO pool minus mean (pool_minus_mean family)	poolq_loo_mean
$C_{i,t}, L_{C,\text{LOO}}$	Viable-peer congestion	crowding_smooth	Pool size / Λ (prose hook)	pool_size_oa_loo (optional)
S_i	Selection / evaluation score	Generative + score modes in CELL 10	— (empirical only v1)	—
Y	Advancement outcome	Y_draft / draft indicator	Promotion / attrition (competing risks)	tenure_event
Λ	Global slot / board capacity	Draft class size (secondary)	Board size, promotion slots	Tenure lines (prose)

§1 — Ontology (one model, two readouts)

1.1 Net local environment — $L_{\text{net}} = B(\cdot) - D(\cdot)$

Context: Master decomposition in VECTOR / COMPASS framing. **Not** the same object as the Alex selection score alone — the score is the **D-leg** entering selection.

$$L_{\text{net}} = B(\cdot) - D(\cdot)$$

Symbol	Meaning
L_{net}	Net effect of the local peer environment on advancement propensity (reduced form)
$B(\cdot)$	Benefit leg — visibility, norms, elite exposure, opportunity (e.g. minutes, development upside)
$D(\cdot)$	Constraint leg — competitive congestion, substitutability, finite slots

Plain English: Being around strong peers can **help** (B) and **hurt** (D). The inverted-U is the empirical signature that both operate.
Sources: #05 §2 · #03 modeling status · #04 §2.
v1 status: Conceptual spine — **not** separately estimated as $B(Q) - D(Q)$ in v1 (deferred).

§2 — Selection score (Alex / Tier 1 generative)

2.1 Canonical Alex score — $S_i = A_i - \lambda L_{C,LOO,i}$
Context: Minimal **generative** selection rule (538D CELL 10). Operationalizes **congestion in who gets picked**, not full L_{net} .

$$S_i = A_i - \lambda L_{C,LOO,i}$$

Symbol	Meaning
S_i	Selection / evaluation score for individual i
A_i	Own latent ability (generative draw; maps to perf / ability in sim)
$L_{C,LOO,i}$	Leave-one-out viable-peer congestion on i 's pool
λ	Weight on congestion penalty (frozen preset ≈ 0.55 in 539 bundle)

Plain English: Advancement ranking = **how good you are** minus **how crowded your viable peers are**.
Repo: sports/538D_development.ipynb CELL 10 · tier1_cell10_playground_state.json · D10 score_equation_one_pager.md .
v1 status: Required generative POC (with ability-only null at $\lambda = 0$).

2.2 Congestion score mode (CELL 10) — $S_i = A_i - w L_C$
Context: Widget label in CELL 10 when score_mode = crowding_smooth . Same idea as §2.1; w is the UI slider for λ .

$$S_i = A_i - w L_C$$

Symbol	Meaning
w	Congestion weight slider (loo_gap_weight in state JSON)
L_C	crowding_smooth — LOO mean of smooth viability (see §3.4)

Sources: 538_Cell10_Generative_Manual.md · #03 (generative row).
2.3 LOO-gap blend mode — $S_i = w(A_i - L_Q) + (1 - w)A_i$
Context: Alternate CELL 10 score — mixes **gap below pool mean** with raw ability.

$$S_i = w (A_i - L_Q) + (1 - w) A_i$$

Symbol	Meaning
L_Q	LOO pool quality (mean teammate ability on roster, excluding self)
w	Blend weight toward LOO gap vs pure ability

Plain English: Penalize being **below** your teammates' average, weighted by w .
Repo: CELL 10 dropdown loo_gap_plus_ability .

2.4 Comparative evaluation (theory prose)
Context: Alex mechanism #1 — evaluators see you **relative to local peers**.

$$\text{signal}_i = a_i - f(\text{local peer quality}) + \epsilon_i$$

Symbol	Meaning
a_i	True ability
$f(\cdot)$	How much local peer caliber shifts perceived signal
ϵ_i	Noise

Plain English: Draft odds depend on **standing out**, not raw stats alone.
Sources: Alex_model_interpreted.md · PD12 comparative-evaluation notes.

§3 — Pool quality & congestion (measurement)

3.1 Team mean ability — $\bar{a}_t = \frac{1}{n_t} \sum_j a_j$
Context: Pool **quality** construct; generative readout axis (**pool mean**).

$$\bar{a}_t = \frac{1}{n_t} \sum_{j=1}^{n_t} a_j$$

Symbol	Meaning
$\bar{a}_t$	Mean ability on team / pool t
n_t	Roster or pool size
a_j	Ability of member j

Empirical proxy (basketball): `poolq_loo` on LOO axis — **not** identical to $\bar{a}_t$ (excludes self; different conditioning).

Sources: `Alex_model_interpreted.md` · `#04` §2.5.

3.2 LOO pool quality — L_Q → `poolq_loo`

Context: Rung 1 empirical x-axis for basketball (and tenure analog).

Definition (basketball): For player i on team-season t ,

$$L_{Q,i} = \frac{1}{n_t - 1} \sum_{j \neq i} \text{perf}_j$$

Symbol	Meaning
$L_{Q,i}$	Leave-one-out mean teammate performance
<code>perf</code>	Chosen performance metric (often PPM z within season)

Army analog: LOO senior-rater **pool minus mean** on talent bucket — column family `pool_minus_mean` / `z_pool_minus_mean_snr_fwd`.

Tenure analog: `poolq_loo_mean` — LOO mean dept peer publication intensity.

Plain English: “How good are my **teammates** (without counting me)?”

Sources: `#01` glossary · `#07` claim table · `530` / `538` empirical ladder.

3.3 Viable-peer congestion (count) — $C_{i,t}$

Context: Alex **crowding** — teammates who **compete for the same scarce slot**, not merely “good teammates.”

$$C_{i,t} = \frac{1}{n_t - 1} \sum_{j \neq i} \mathbf{1}(a_j > \theta)$$

Symbol	Meaning
$C_{i,t}$	Share/count of teammates above viability threshold θ
θ	Prospect threshold (PPM quantile, draft signal, combine, etc.)
$\mathbf{1}(\cdot)$	Indicator — 1 if viable peer, else 0

Role-specific variant: $C_{i,t}^{\text{role}} = \frac{1}{n_t - 1} \sum_{j \neq i} w_{ij} \mathbf{1}(a_j > \theta)$ with position weights w_{ij} .

Sources: `Alex_model_interpreted.md` §NCAA operationalization.

3.4 Smooth viability & `crowding_smooth`

Context: Implemented congestion in CELL 10 / 530 export (539 preset).

$$v_j = \sigma(\gamma(a_j - \theta)), \quad L_{C,\text{LOO},i} = \frac{1}{n_t - 1} \sum_{j \neq i} v_j$$

Symbol	Meaning
v_j	Smooth viability of teammate j
$\sigma(\cdot)$	Logistic sigmoid
γ	Sharpness (preset ≈ 10)
θ	Viability midpoint (preset ≈ 0.72 on $[0,1]$ scale)
$L_{C,\text{LOO},i}$	LOO mean viability → column <code>crowding_smooth</code>

Plain English: Congestion = **how many plausible rivals** sit on your roster, smoothed so counts aren’t brittle.

Repo: `tier1_pool_assignment.py` · `SELECTION_539_*` in `tier1_sim_config.py`.

Do not confuse: `pool_c_loo` (raw LOO sum) $\approx (n - 1) \times$ mean at fixed n — legacy; prefer `crowding_smooth`.

3.5 Quality vs congestion (Rung 2.5)

Context: PD12 Priority 3 — theory proposes **two measurements**, not one blended neighbor stat.

Construct	Symbol / column	Leg	Question it answers
Team quality	<code>poolq_loo</code> , $\bar{a}_t$	B–D mixed (diagnostic)	“How strong is my peer group?”
Congestion	<code>crowding_smooth</code> , $C_{i,t}$	D	“How many rivals for the same slot?”

Sources: `#04` §2.5 · `#05` §5 · `#12` §2.4.

§4 — Empirical inverted-U ladder (basketball / tenure sign-checks)

4.1 Ventile bins (descriptive)

Context: Rung 1 — see the curve before parametric fit.
Procedure: Sort individuals by poolq_loo into ventiles (or equal-width bins); plot mean draft (or tenure) rate vs bin midpoint.

Object	Meaning
Bin midpoint	Mean poolq_loo in bin
Bin outcome	Mean Y_draft or share drafted

Repo: 530_sports_pipeline.ipynb · 538 CELL 4–6 · #08 figure list.

4.2 Linear probability model (LPM)

Context: Shape diagnostic — quadratic curvature for inverted-U.

Y_i = beta_0 + beta_1 L_{Q,i} + beta_2 L_{Q,i}^2 + (controls) + epsilon_i

Symbol	Meaning
Y_i	Binary outcome (drafted, tenured, promoted)
L_{Q,i}	poolq_loo (or domain analog)
beta_2 < 0	Concave — inverted-U if beta_1 > 0 at interior

Plain English: OLS on 0/1 — coefficients ≈ percentage-point changes; use for shape, not final inference.
Sources: #01 LPM glossary · #03 empirical ladder row · 538 CELL 5.

4.3 Logit (proper binary link)

Context: Same predictors as LPM, logistic link for probabilities in (0, 1).

Pr(Y_i = 1) = logit^{-1}(beta_0 + beta_1 L_{Q,i} + beta_2 L_{Q,i}^2 + ...)

Repo: 538 empirical ladder step after LPM.

4.4 Peak location — L* (L-star)

Context: Where the quadratic LPM predicts maximum expected outcome.

L* = -beta_1 / (2*beta_2) (beta_2 != 0)

Symbol	Meaning
L*	Pool-quality level at predicted peak of inverted-U

Plain English: “At what teammate-quality level does draft probability top out?”
Sources: #01 · #03 · 538 CELL 5 (Alex §6 step 2).
v1 status: Beneficial to report all three settings; not all exported yet.

§5 — Alex empirical draft models (mediation & heterogeneity)

5.1 Baseline draft hazard — quadratic in team quality

P(draft_i) = f(a_i, a_bar_t, a_bar_t^2)

Predictions: dP/d a_i > 0; team quality initially positive; quadratic term captures downturn at elite a_bar_t.
Sources: Alex_model_interpreted.md .

5.2 Mechanism regression — add congestion

Y_i ~ a_i + a_bar_t + a_bar_t^2 + C_{i,t}

Plain English: If C_{i,t} is negative and shrinks the negative quadratic, congestion mediates the elite-pool dip.

5.3 Near-threshold interaction (Prediction #1)

Y_i ~ a_i + a_bar_t + C_{i,t} + a_i x C_{i,t}

Plain English: Congestion hurts most for borderline talents (high a_i but not superstar), not the very best or weak.
Repo: 538D CELL 4D → heterogeneity_ventiles_top_tail.png (#08).

5.4 Assortativity interaction (exploratory)

Y_i ~ a_i + a_bar_t + a_bar_t^2 + A_s . a_bar_t^2

Symbol	Meaning
A_s	System assortativity (how strongly talent sorts into elite pools)

Plain English: Stronger sorting → stronger right-tail penalty in team quality.

5.5 Mean × SD peer dispersion (538D 4B/4C — exploratory)

Context: Downturn may depend on **spread** of talent within pool, not mean alone.

Idea: Interact or joint-model `poolq_loo` with `peer_perf_sd_loo` (LOO SD of teammate `perf`).

v1 status: Exploratory EDA — supplement tier, not closure blocker.

§6 — Generative simulation (CELL 10)

Repo (this section): `sports/538D_development.ipynb` CELL 10 — primary generative lab for v1 closure. Same CELL 10 playground also lives in `sports/538_alex_tier1_model_and_fit.ipynb` (empirical + generative in one notebook). Operator manual: `sports/documents/538_Cell10_Generative_Manual.md` . Code: `sports/tier1_cell10_playground_run.py` (widget UI), `sports/tier1_pool_assignment.py` (soft assign), `sports/tier1_generative_eda.py` (Plot B). State file: `sports/tier1_cell10_playground_state.json` . Sort-and-chop null benchmark: `sports/537_tier1_benchmark.ipynb` (537 B overlay).

6.1 Soft assignment kernel

Context: Realistic **overlapping** pools (not sort-and-chop). Player i joins team j with weight:

$$\tilde{\pi}_{ij} = f(A_i - T_j) \cdot (n_j + k)^\alpha$$

Symbol	Meaning
T_j	Fixed target mean for team j (drawn once)
τ	Temperature — scales $(A_i - T_j)$ in kernel
f	Gaussian $\exp(-(A_i - T_j)^2/(2\tau^2))$ or Cauchy $1/(1 + ((A_i - T_j)/\tau)^2)$
α	Preferential attachment — rich teams attract more players
n_j	Current roster count during sequential placement

Plain English: Players land on teams whose **target level** matches their ability, with noise and overlap.

Repo: `sports/tier1_pool_assignment.py` · `sports/537_tier1_benchmark.ipynb` falsifies disjoint sort-and-chop.

6.2 Selection rule — top- K

Context: After assignment, highest S_i get `Y_selected = 1` (draft/promotion slots).

Symbol	Meaning
K	<code>N_SELECTED</code> — number of advancement slots
N	Total synthetic players ($J \times$ roster size)

Plot B readout: Mean selection rate vs bins of **x-axis** (see §9).

6.3 Ability-only null

$$S_i = A_i \quad (w = 0, \lambda = 0)$$

Context: Fails to produce inverted-U alone — required closure check (SCOUT C1).

§7 — Survival & time-to-event (Army, tenure)

7.1 Cox proportional hazards (generic)

$$h_i(t) = h_0(t) \exp(\beta_1 x_{i1} + \beta_2 x_{i2} + \cdots)$$

Context: Army cause-specific Cox; tenure Layer B (pre-submission).

Symbol	Meaning
$h_i(t)$	Hazard of event at time t for person i
$h_0(t)$	Baseline hazard
x_{ik}	Covariates (often includes LOO pool quality + quadratic)

7.2 Alex delay / queueing prediction

$$h_i(t) = h_0(t) \exp(\beta_1 a_i + \beta_2 \bar{a}_t + \beta_3 C_{i,t}), \quad \beta_3 < 0 \text{ for high } a_i$$

Plain English: Congestion delays advancement, not only binary failure.

Army v1: CIF bar panels on LOO pool-minus-mean bins + cause-specific Cox with quadratics — **not** Fine–Gray in current stack (`#07`).

7.3 Tenure Cox covariates (planned Layer B)
Typical specification (from PEER correspondence):

$$\text{tenure_hazard} \sim z_pool_minus_mean_snr_fwd + z_pool_minus_mean_snr_fwd_sq + \text{controls}$$

Column	Meaning
z_pool_minus_mean_snr_fwd	Z-scored LOO dept peer quality
..._sq	Quadratic — inverted-U test

v1 status: Stage 9 binned plot on poolq_loo_mean is preliminary; Cox not archived for draft.

7.4 CIF (cumulative incidence)

Context: Army Cell 11 descriptive bar panels — empirical within-bin cumulative incidence, not Cox-predicted curves (#07 guardrail).

Plain English: “Of people in this pool-quality bin, what fraction eventually promoted / attrited?”

§8 — Prediction hooks (variables only)

Prediction	Formula hook	Key variables
#1 Near-threshold	§5.3 interaction	$a_i \times C_{i,t}$ or ventile slices by own perf
#2 Peak shift with Λ	Move L^* or peak bin with global slots	Λ = board size, draft class size, tenure lines
Mean × SD dispersion	§5.5	peer_perf_sd_loo × poolq_loo
Own-TB stratified U (Army)	Stratify CIF/Cox by own talent bucket	Own TB × pool quality
Assortativity for U	§6.1 soft assign vs §6.3 null	Overlap + congestion required

Sources: #03 Predictions section · #04 §3.

§9 — Axis discipline (mandatory honesty)

Context: The #1 source of over-claiming in v1.

Readout	Typical x-axis	Claim level
Empirical inverted-U (Rung 1)	LOO pool quality (poolq_loo , pool_minus_mean , poolq_loo_mean)	Supported / preliminary by domain
Generative POC (Rung 2)	Pool mean (team_mean)	Qualitative peak-and-decline only
Same generative score on LOO axis	poolq_loo bins	Mostly decreasing — limitation row

Frozen sentence (paste into manuscript): see #05 §4 and #12 §3.

Artifact: D10 axis_table_generative_readouts.md — explicit row-per-quantity table (#03 “explicit axis table”).

§10 — Explicitly deferred (named so you don’t hunt)

Formula / object	Why deferred
Full $B(Q) - D(Q)$ generative decomposition	Not estimated v1
LOO generative bin-for-bin match	Path A — parallel north star only
Multiplicative talent production (Menger/Shockley)	Manuscript discussion, not v1 mechanism
Fine–Gray subdistribution hazards	Army — revisit pre-publication
3-domain parametric identifiability	Post-draft (PD12 P1)
Evans et al. full sim embed	Theory reference only

Sources: #05 §7 · #03 Part III defer rows.

Source index (where formulas were gleaned)

Document	Role
03_Where_we_are_now.md	Status + equation-level claim S_i
04_Project_story_plain_English.md	Plain construct map
05_Model_Nesting_Note_v1.md	Canonical ontology + column map
06_Generative_closure_checklist.md	What must be true for closure
07_Claim_language_guardrails.md	What you may say per formula readout
12_Manuscript_staging_prose.md	Ink-ready generative sentence
Alex_model_interpreted.md	Full Alex mechanism formulas
538_Cell10_Generative_Manual.md	CELL 10 symbols + score modes
sports/scripts/export_scout_manuscript_bundle_v1.py	D10 axis table + score one-pager template

When you finish #05, treat this file as the **fast index**; #05 remains the frozen nesting authority for manuscript §5.